ScholarPlan: E-Portfolio Supporting Content

This document contains:

- *Critical evaluation*
- *Skill Matrix*
- *Three formative case studies*

1. Critical Evaluation of the ScholarPlan Project

ScholarPlan is an LLM-powered planning agent designed to support academic research and information gathering. The system separates the workflow into five specialised roles: Planner, Retriever, Processor, Critic and Archivist. A SQLite blackboard provides shared state so that agents can exchange research goals, retrieved sources, claims and execution information. The design was developed collaboratively in Week 7, while I implemented the system individually in Week 11. This distinction is important because the final implementation required me to translate a shared conceptual design into working code and test its components.

A key strength is the modular architecture. Separating planning, retrieval, processing, verification and storage makes responsibilities easier to understand and components easier to test independently. The Planner can decompose a research goal and support replanning; the Retriever obtains scholarly material through sources such as arXiv, OpenAlex and Crossref; the Processor summarises information; the Critic independently checks generated claims; and the Archivist stores outputs and execution traces. The blackboard offers a clear coordination mechanism without requiring every agent to communicate directly with every other agent. This is a practical architectural choice for a system whose intermediate results need to remain visible and traceable.

The implementation and testing process also exposed the difference between a plausible design and a reliable system. LLM outputs are not fully deterministic, scholarly APIs can impose rate limits or return incomplete results, and an agent may need to recover when a step fails or evidence is insufficient. These constraints mean that a successful run alone is not proof of overall reliability. The recorded test run showed 24 tests passing across the agent loop, blackboard, Critic, Planner, Processor, Retriever, Archivist and evaluation metrics. This is useful evidence of component-level behaviour, but it does not establish research quality across a broad range of real user queries.

The tutor’s feedback recognised the value of the architecture, the SQLite blackboard, the Critic verifier and the discussion of cost, API limits and nondeterminism. It also identified specific improvements: a UML class diagram; measurable evaluation criteria, including test queries, deduplication thresholds and Critic precision/recall; and explicit fallback rules when

repeated verification or replanning fails. These are substantive limitations rather than merely presentation changes. Without defined thresholds and a stopping rule, the system may behave inconsistently or spend excessive time and API calls trying to resolve a conflict.

My overall judgement is that ScholarPlan demonstrates a coherent prototype architecture and a useful separation of agent responsibilities, but it should not yet be treated as an autonomous authority on academic evidence. The next iteration should use a fixed evaluation set, measure retrieval relevance and claim-verification performance, test duplicate detection, and record latency and cost. It should also cap replanning attempts—for example, at three—before returning an explicit unresolved status for human review. This would make success criteria reproducible and make failure safer and easier to explain.

2. Skill Matrix



3. Case Study: Bias in Autonomous Decision-Making

The first formative case study considers how bias can enter an autonomous research workflow. ScholarPlan retrieves and processes scholarly material, but its outputs depend on what is available from the selected APIs, how the retrieval query is formulated, and which sources the LLM prioritises or summarises. If the retrieved set overrepresents a particular discipline, geography, language or viewpoint, the resulting synthesis may reproduce that imbalance even if each individual source is genuine.

The multi-agent design helps organise the workflow, but modularity does not automatically remove bias. The Planner may frame the research question too narrowly, the Retriever may return uneven coverage, and the Processor may give more prominence to claims that are repeated or expressed confidently. The Critic can challenge unsupported claims, yet it may share some of the same model limitations or rely on the same evidence pool. Independent verification therefore reduces some risks but cannot guarantee neutrality.

A practical improvement would be to test ScholarPlan against a balanced set of research questions from different domains and perspectives. For each query, I would record source coverage, publication dates, disciplinary distribution and whether credible counterarguments were represented. Retrieval quality should be assessed separately from the quality of the final summary. Where coverage is narrow, the system should state that limitation rather than imply that the retrieved sources represent the whole field. A human reviewer should also be able to inspect the sources and challenge the system's framing.

This case study changed how I interpret an agent's apparently objective output. A structured workflow can make decisions more consistent, but it can also make hidden assumptions harder to notice if users trust the final answer without inspecting the evidence. I would therefore treat bias analysis as a design and evaluation requirement, not as a final disclaimer.

4. Case Study: AI Safety Constraints and Failure Recovery

The second case study focuses on safety and control. ScholarPlan can decompose a goal, retrieve information, generate claims and re-plan when a step is unsuccessful. These capabilities improve flexibility, but autonomous iteration introduces risks: repeated retries can increase cost, contradictory evidence may remain unresolved, and an LLM may produce a plausible answer despite inadequate support. External APIs may also fail or return incomplete results.

The blackboard and execution traces provide useful control points because they make intermediate state and previous actions inspectable. The Critic adds a verification stage before claims are accepted. However, safety depends on the system's response when verification fails. If the agent simply retries indefinitely or silently accepts a weak claim, the architecture's safeguards are insufficient. Tutor feedback specifically highlighted the need for explicit fallback rules for repeated conflicts or rejection.

In a future version, I would define a maximum of three replanning attempts for a given unresolved step, apply timeouts and retry limits to external API calls, and distinguish verified, partially supported and unresolved claims. If the Critic cannot establish adequate support, ScholarPlan should preserve the uncertainty and return the issue for human review rather

than inventing a resolution. Tests should cover missing API responses, conflicting sources, empty retrieval results, repeated Critic rejection and exhausted retry limits.

This case study reinforced that autonomy should be bounded by clear policies. A system is not safer simply because it has more agents or more opportunities to retry. Safety comes from explicit limits, observable state, predictable failure handling and a clear handover to a person when the system reaches the boundary of what it can establish.

5. Case Study: Explainability and Trust

The third case study examines explainability. In an academic research assistant, a user needs more than a fluent summary: they need to understand which sources support a claim, what the system did to obtain the result and where uncertainty remains. ScholarPlan's modular roles and SQLite blackboard create a basis for this transparency. Execution traces can record the steps taken, while the Critic can document why a claim was accepted, challenged or left unresolved.

Nevertheless, a trace of system activity is not the same as a complete explanation of a model's internal reasoning. It can show which query was issued, which sources were retrieved and which checks were applied, but it cannot guarantee that an LLM's generated rationale faithfully represents every factor that influenced its output. Explainability should therefore focus on evidence and decision-relevant information rather than presenting generated reasoning as a definitive account of internal cognition.

I would improve the user-facing output by linking each important claim to its source, displaying verification status, identifying disagreements between sources and recording why a claim could not be verified. The interface should separate source evidence from the system's own synthesis and include a concise limitations section. These features could be evaluated through user tasks—for example, asking reviewers to trace claims to sources, identify unsupported statements and judge whether uncertainty is communicated clearly.

This case study helped me distinguish transparency from trustworthiness. A system may be easy to inspect yet still make mistakes; conversely, a correct answer may be difficult to audit if its evidence trail is missing. For ScholarPlan, explainability should make it easier for a researcher to verify the output independently, rather than asking the researcher to trust the agent.

References

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